

Solutions for Practice Examination 2
Mathematical Probability (STAT2003/STAT7003)
Created during Semester 1, 2026

Question 1. [10 marks]

Let $X_1, X_2, X_3 \stackrel{\text{iid}}{\sim} \text{Exp}(\lambda)$ with $\lambda > 0$ (and PDF $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$, $f(x) = 0$ for $x < 0$). Denote $X_{(1)} = \min(X_1, X_2, X_3)$ and $X_{(3)} = \max(X_1, X_2, X_3)$.

- (a) Find the CDF and PDF of $X_{(1)}$, and identify the resulting distribution by name and parameters (e.g. $\text{Exp}(\cdot)$, $\text{Gamma}(\cdot, \cdot)$, $\mathcal{U}(\cdot, \cdot)$, etc.).

[3 marks]

- (b) Compute $\mathbb{E}[X_{(1)}]$.

[1 mark]

- (c) Find the CDF of $X_{(3)}$ and compute $\mathbb{P}\left(X_{(3)} > \frac{2}{\lambda}\right)$.

[3 marks]

- (d) Using the identity $X_{(1)} + X_{(2)} + X_{(3)} = X_1 + X_2 + X_3$, compute $\mathbb{E}[X_{(3)}]$ given that $\mathbb{E}[X_{(2)}] = \frac{5}{6\lambda}$.

[3 marks]

Solution:

- (a) $\mathbb{P}(X_{(1)} > x) = \mathbb{P}(X_1 > x, X_2 > x, X_3 > x) = (e^{-\lambda x})^3 = e^{-3\lambda x}$.

$$F_{X_{(1)}}(x) = 1 - e^{-3\lambda x}, \quad x \geq 0.$$

$$f_{X_{(1)}}(x) = 3\lambda e^{-3\lambda x}, \quad x \geq 0.$$

This is $\text{Exp}(3\lambda)$: the minimum of 3 iid exponentials with rate λ is exponential with rate 3λ .

- (b) $\mathbb{E}[X_{(1)}] = \frac{1}{3\lambda}$.

- (c) $F_{X_{(3)}}(x) = \mathbb{P}(X_1 \leq x, X_2 \leq x, X_3 \leq x) = (1 - e^{-\lambda x})^3, \quad x \geq 0.$

$$\mathbb{P}\left(X_{(3)} > \frac{2}{\lambda}\right) = 1 - F_{X_{(3)}}\left(\frac{2}{\lambda}\right) = 1 - (1 - e^{-2})^3 \approx 1 - 0.6465 = 0.3535.$$

- (d) Since $X_{(1)} + X_{(2)} + X_{(3)} = X_1 + X_2 + X_3$, taking expectations:

$$\mathbb{E}[X_{(1)}] + \mathbb{E}[X_{(2)}] + \mathbb{E}[X_{(3)}] = \mathbb{E}[X_1] + \mathbb{E}[X_2] + \mathbb{E}[X_3] = \frac{3}{\lambda}.$$

(Note: the given value $\mathbb{E}[X_{(2)}] = 5/(6\lambda)$ can be derived from the order statistic PDF. For $n = 3, k = 2$: $f_{X_{(2)}}(x) = 6F(x)(1 - F(x))f(x) = 6\lambda(e^{-2\lambda x} - e^{-3\lambda x})$, so $\mathbb{E}[X_{(2)}] = 6\lambda \left[\frac{1}{(2\lambda)^2} - \frac{1}{(3\lambda)^2} \right] = 6\lambda \cdot \frac{5}{36\lambda^2} = \frac{5}{6\lambda}$.)

$$\mathbb{E}[X_{(3)}] = \frac{3}{\lambda} - \frac{1}{3\lambda} - \frac{5}{6\lambda} = \frac{18 - 2 - 5}{6\lambda} = \frac{11}{6\lambda}.$$

$$\text{Direct check: } \mathbb{E}[X_{(3)}] = \int_0^\infty (1 - F_{X_{(3)}}(x)) dx = \int_0^\infty (3e^{-\lambda x} - 3e^{-2\lambda x} + e^{-3\lambda x}) dx = \frac{3}{\lambda} - \frac{3}{2\lambda} + \frac{1}{3\lambda} = \frac{18 - 9 + 2}{6\lambda} = \frac{11}{6\lambda}. \checkmark$$

Question 2. [10 marks]

Consider the following Python program:

```
def experiment():
    u = random()
    v = random()
    if u < 0.5:
        if v < 0.4:
            return 0
        else:
            return 1
    else:
        if v < 0.6:
            return 1
        else:
            return 2
```

Let X denote the value returned by `experiment()`.

- (a) X is a random variable. Find the PMF of X , carefully justifying how the two calls to `random()` combine to produce each outcome.

[3 marks]

- (b) Compute $\mathbb{E}[X]$ and $\text{Var}(X)$.

[2 marks]

- (c) Write down the MGF $M_X(t) = \mathbb{E}[e^{tX}]$ and use it to confirm that $M'_X(0) = \mathbb{E}[X]$.

[2 marks]

- (d) Consider the following code:

```
s = 0
for _ in range(100):
    s = s + experiment()
print(s)
```

The output is a random variable S . Using the CLT, express $\mathbb{P}(S > 110)$ in terms of Φ , the standard normal CDF.

[3 marks]

Notes about Python:

- `random()` generates a $\mathcal{U}(0, 1)$ random variable.

Solution:

(a) The program uses two independent $U(0, 1)$ draws u and v .

$\mathbb{P}(X = 0)$: requires $u < 0.5$ and $v < 0.4$, so $\mathbb{P}(X = 0) = 0.5 \times 0.4 = 0.20$.

$\mathbb{P}(X = 1)$: either $(u < 0.5 \text{ and } v \geq 0.4)$ or $(u \geq 0.5 \text{ and } v < 0.6)$, so $\mathbb{P}(X = 1) = 0.5 \times 0.6 + 0.5 \times 0.6 = 0.60$.

$\mathbb{P}(X = 2)$: requires $u \geq 0.5$ and $v \geq 0.6$, so $\mathbb{P}(X = 2) = 0.5 \times 0.4 = 0.20$.

So $\mathbb{P}(X = 0) = 0.2$, $\mathbb{P}(X = 1) = 0.6$, $\mathbb{P}(X = 2) = 0.2$.

(b) $\mathbb{E}[X] = 0(0.2) + 1(0.6) + 2(0.2) = 0 + 0.6 + 0.4 = 1.0$.

$\mathbb{E}[X^2] = 0(0.2) + 1(0.6) + 4(0.2) = 0 + 0.6 + 0.8 = 1.4$.

$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = 1.4 - 1.0 = 0.4$.

(c) $M_X(t) = 0.2 e^0 + 0.6 e^t + 0.2 e^{2t} = 0.2 + 0.6 e^t + 0.2 e^{2t}$.

$M'_X(t) = 0.6 e^t + 0.4 e^{2t}$, so $M'_X(0) = 0.6 + 0.4 = 1.0 = \mathbb{E}[X]$. □

(d) $S = X_1 + X_2 + \cdots + X_{100}$ where X_i are iid with $\mathbb{E}[X_i] = 1$ and $\text{Var}(X_i) = 0.4$.

$\mathbb{E}[S] = 100 \times 1 = 100$. $\text{Var}(S) = 100 \times 0.4 = 40$. $\text{SD}(S) = \sqrt{40} = 2\sqrt{10}$.

By the CLT, $S \sim \mathcal{N}(100, 40)$, so

$\mathbb{P}(S > 110) \approx \mathbb{P}\left(Z > \frac{110 - 100}{2\sqrt{10}}\right) = \mathbb{P}\left(Z > \frac{5}{\sqrt{10}}\right) = 1 - \Phi\left(\frac{5}{\sqrt{10}}\right) = 1 - \Phi(\sqrt{2.5}) \approx 1 - \Phi(1.581) \approx 0.057$.

Question 3. [10 marks]

The weather on day n is in one of three states: Clear (1), Cloudy (2), or Overcast (3). Each day, the volume of clouds V (in arbitrary units) depends on the current state and follows an exponential distribution:

- State 1 (Clear): $V \sim \text{Exp}(1)$ (mean 1).
- State 2 (Cloudy): $V \sim \text{Exp}(1/3)$ (mean 3).
- State 3 (Overcast): $V \sim \text{Exp}(1/5)$ (mean 5).

Let m_i denote the median of V in state i , and let $Q_1^{(i)}$ and $Q_3^{(i)}$ denote the first and third quartiles (i.e. the 0.25 and 0.75 quantiles) of V in state i . The transition rules are:

- From state 1: if $V > m_1$, move to state 2; otherwise stay in state 1.
 - From state 2: if $V < Q_1^{(2)}$, move to state 1; if $V > Q_3^{(2)}$, move to state 3; otherwise stay in state 2.
 - From state 3: if $V \leq m_3$, move to state 2; otherwise stay in state 3.
- (a) For an $\text{Exp}(\lambda)$ random variable, show that the median is $\ln(2)/\lambda$. Find expressions for Q_1 and Q_3 in terms of λ . [2 marks]
- (b) Using the transition rules, write down the 3×3 transition matrix $\mathbf{P}$ for the weather Markov chain. [2 marks]
- (c) Find the stationary distribution $\boldsymbol{\pi}$ by solving $\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}$. [3 marks]
- (d) In the long run, what is the expected daily cloud volume? Use the law of total expectation conditioning on the state. [3 marks]

Solution:

- (a) For $V \sim \text{Exp}(\lambda)$, the CDF is $F(v) = 1 - e^{-\lambda v}$. The p -quantile q_p satisfies $F(q_p) = p$:
- $$1 - e^{-\lambda q_p} = p \implies q_p = -\ln(1 - p)/\lambda.$$
- Median ($p = 1/2$): $m = \ln(2)/\lambda$. Then $\mathbb{P}(V > m) = 1/2$.
- Q_1 ($p = 1/4$): $Q_1 = -\ln(3/4)/\lambda = \ln(4/3)/\lambda$.
- Q_3 ($p = 3/4$): $Q_3 = -\ln(1/4)/\lambda = \ln(4)/\lambda$.
- Note: $\mathbb{P}(V < Q_1) = 1/4$, $\mathbb{P}(V > Q_3) = 1/4$, $\mathbb{P}(Q_1 \leq V \leq Q_3) = 1/2$.
- (b) From state 1: $\mathbb{P}(V > m_1) = 1/2$, so stay with prob $1/2$, move to 2 with prob $1/2$.
- From state 2: $\mathbb{P}(V < Q_1^{(2)}) = 1/4$ (move to 1), $\mathbb{P}(V > Q_3^{(2)}) = 1/4$ (move to 3), $\mathbb{P}(Q_1^{(2)} \leq V \leq Q_3^{(2)}) = 1/2$ (stay).
- From state 3: $\mathbb{P}(V \leq m_3) = 1/2$, so move to 2 with prob $1/2$, stay with prob $1/2$.

$$\mathbf{P} = \begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/4 & 1/2 & 1/4 \\ 0 & 1/2 & 1/2 \end{pmatrix}$$

(c) Solve $\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}$:

$$\pi_1 = (1/2)\pi_1 + (1/4)\pi_2 \implies (1/2)\pi_1 = (1/4)\pi_2 \implies \pi_1 = (1/2)\pi_2.$$

$$\pi_3 = (1/4)\pi_2 + (1/2)\pi_3 \implies (1/2)\pi_3 = (1/4)\pi_2 \implies \pi_3 = (1/2)\pi_2.$$

So $\pi_1 = \pi_3$. Normalise: $(1/2)\pi_2 + \pi_2 + (1/2)\pi_2 = 2\pi_2 = 1$, so $\pi_2 = 1/2$.

$$\boldsymbol{\pi} = (1/4, 1/2, 1/4).$$

The weather is Cloudy half the time, and Clear or Overcast each a quarter of the time.

(d) $\mathbb{E}[V] = \pi_1 \cdot \mathbb{E}[V \mid \text{state 1}] + \pi_2 \cdot \mathbb{E}[V \mid \text{state 2}] + \pi_3 \cdot \mathbb{E}[V \mid \text{state 3}]$

$$= \frac{1}{4} \cdot 1 + \frac{1}{2} \cdot 3 + \frac{1}{4} \cdot 5 = \frac{1 + 6 + 5}{4} = 3.$$

Question 4. [10 marks]

Let (X, Y) have a bivariate normal distribution with $\mu_X = 1$, $\mu_Y = 4$, $\sigma_X^2 = 9$, $\sigma_Y^2 = 16$, and $\rho = -0.5$.

- (a) Compute $\mathbb{P}(2X - Y > 1)$, expressing your answer in terms of Φ .

[3 marks]

- (b) Find the conditional distribution of Y given $X = 4$, and compute $\mathbb{P}(Y > 6 | X = 4)$.

[3 marks]

- (c) Let $A = X + Y$ and $B = X - Y$. Determine the value of ρ (the original correlation) that would make A and B independent. With the given $\rho = -0.5$, compute $\text{Cov}(A, B)$ and state whether A and B are independent.

[2 marks]

- (d) Define $C = X + 2Y$ and $D = 3X + aY$ for some constant a . Find a such that C and D are independent.

[2 marks]

Solution:

- (a) Let $W = 2X - Y$. Since (X, Y) is bivariate normal, any linear combination is normal.

$$\mathbb{E}[W] = 2(1) - 4 = -2.$$

$$\text{Var}(W) = 4\sigma_X^2 + \sigma_Y^2 - 2(2)(1)\text{Cov}(X, Y), \text{ where } \text{Cov}(X, Y) = \rho \sigma_X \sigma_Y = (-0.5)(3)(4) = -6.$$

$$\text{Var}(W) = 4(9) + 16 - 4(-6) = 36 + 16 + 24 = 76.$$

$$\mathbb{P}(W > 1) = \mathbb{P}\left(Z > \frac{1 - (-2)}{\sqrt{76}}\right) = \mathbb{P}\left(Z > \frac{3}{\sqrt{76}}\right) = 1 - \Phi\left(\frac{3}{\sqrt{76}}\right) \approx 1 - \Phi(0.344) \approx 0.365.$$

- (b) $\mathbb{E}[Y | X = 4] = \mu_Y + \rho \frac{\sigma_Y}{\sigma_X}(4 - \mu_X) = 4 + (-0.5)\left(\frac{4}{3}\right)(3) = 4 - 2 = 2.$

$$\text{Var}(Y | X) = \sigma_Y^2(1 - \rho^2) = 16(1 - 0.25) = 12.$$

$$\text{So } Y | X = 4 \sim \mathcal{N}(2, 12).$$

$$\mathbb{P}(Y > 6 | X = 4) = \mathbb{P}\left(Z > \frac{6 - 2}{\sqrt{12}}\right) = \mathbb{P}\left(Z > \frac{2}{\sqrt{3}}\right) = 1 - \Phi(1.155) \approx 0.124.$$

- (c) $\text{Cov}(A, B) = \text{Cov}(X + Y, X - Y) = \text{Var}(X) - \text{Var}(Y) + \text{Cov}(Y, X) - \text{Cov}(X, Y) = \sigma_X^2 - \sigma_Y^2 = 9 - 16 = -7.$

Note: the $\text{Cov}(Y, X) - \text{Cov}(X, Y)$ terms cancel. So $\text{Cov}(A, B) = \sigma_X^2 - \sigma_Y^2$ regardless of ρ .

For independence (since (A, B) is jointly normal, being a linear transformation of (X, Y)), we need $\text{Cov}(A, B) = 0$, i.e. $\sigma_X^2 = \sigma_Y^2$. Since $9 \neq 16$, no value of ρ makes A and B independent.

With $\rho = -0.5$: $\text{Cov}(A, B) = -7 \neq 0$, so A and B are **not independent**.

- (d) Since (X, Y) is bivariate normal, (C, D) is jointly normal. Independence $\iff \text{Cov}(C, D) = 0$.

$$\text{Cov}(C, D) = \text{Cov}(X + 2Y, 3X + aY) = 3\text{Var}(X) + a\text{Cov}(X, Y) + 6\text{Cov}(Y, X) + 2a\text{Var}(Y) = 3(9) + a(-6) + 6(-6) + 2a(16) = 27 - 6a - 36 + 32a = 26a - 9.$$

$$\text{Setting } \text{Cov}(C, D) = 0: 26a = 9, \text{ so } a = 9/26.$$

Question 5. [10 marks]

Consider the following Python program:

```
n = 10
while True:
    u = random()
    if u < 0.15:
        print(n)
        break
    n = n + 1
```

Let N be the random variable representing the value of n printed.

(a) Determine $\mathbb{E}[N]$.

[2 marks]

(b) Find the PGF $G_N(z) = \mathbb{E}[z^N]$ and use it to confirm that $G'_N(1) = \mathbb{E}[N]$.

[3 marks]

(c) Find $\mathbb{P}(N > k)$ for $k \geq 10$. Then show that for any integers $s \geq 10$ and $t \geq 1$, $\mathbb{P}(N > s + t | N > s) = \mathbb{P}(N > s + t) / \mathbb{P}(N > s)$ depends only on t .

[3 marks]

(d) Using the result from (c), compute $\mathbb{E}[N | N > 15]$.

[2 marks]

Notes about Python:

- `random()` generates a $\mathcal{U}(0, 1)$ random variable.
- `break` exits the `while` loop immediately.

Solution:

(a) At each iteration, the loop exits with probability $p = 0.15$ or continues with probability 0.85 , and n starts at 10 . So $\mathbb{P}(N = k) = (0.85)^{k-10}(0.15)$ for $k = 10, 11, 12, \dots$

$N = 9 + W$ where $W \sim \text{Geom}(0.15)$ (support $\{1, 2, \dots\}$). So $\mathbb{E}[N] = 9 + 1/0.15 = 9 + 20/3 = 47/3 \approx 15.67$.

(b) Since $N = 9 + W$ where $W \sim \text{Geom}(0.15)$:

$$G_N(z) = \mathbb{E}[z^N] = z^9 \mathbb{E}[z^W] = z^9 \cdot \frac{pz}{1 - (1-p)z} = \frac{pz^{10}}{1 - (1-p)z}.$$

Using the quotient rule:

$$G'_N(z) = \frac{10pz^9(1-qz) + pz^{10}q}{(1-qz)^2} = \frac{pz^9[10(1-qz) + qz]}{(1-qz)^2},$$

where $q = 1 - p = 0.85$. At $z = 1$:

$$G'_N(1) = \frac{p[10p + q]}{p^2} = \frac{10p + q}{p} = 10 + \frac{q}{p} = 10 + \frac{0.85}{0.15} = 10 + \frac{17}{3} = \frac{47}{3} = \mathbb{E}[N]. \quad \checkmark$$

- (c) $\mathbb{P}(N > k) = \mathbb{P}(\text{first } k - 9 \text{ trials all fail}) = (1 - p)^{k-9} = (0.85)^{k-9}$ for $k \geq 9$ (and $\mathbb{P}(N > k) = 1$ for $k < 10$).

For $s \geq 10$ and $t \geq 1$:

$$\mathbb{P}(N > s + t | N > s) = \frac{\mathbb{P}(N > s + t)}{\mathbb{P}(N > s)} = \frac{(0.85)^{s+t-9}}{(0.85)^{s-9}} = (0.85)^t.$$

This depends only on t , not on s (the memoryless property).

- (d) From (c), given $N > 15$, the “remaining” value $N - 15$ behaves like a fresh geometric count starting from 0. So

$$\mathbb{E}[N | N > 15] = 15 + \mathbb{E}[W] = 15 + \frac{1}{0.15} = 15 + \frac{20}{3} = \frac{65}{3} \approx 21.67.$$

(Here $W \sim \text{Geom}(0.15)$ represents the additional trials beyond 15.)

Question 6. [10 marks]

Consider a population model in which each individual independently produces offspring according to: $\mathbb{P}(X = 0) = 0.5$, $\mathbb{P}(X = 1) = 0.3$, $\mathbb{P}(X = 2) = 0.2$. The process starts with one individual ($Z_0 = 1$), and Z_n denotes the population size at generation n . The following Python code simulates the population over g generations:

```
def simulate(g):
    pop = 1
    for gen in range(g):
        new_pop = 0
        for _ in range(pop):
            u = random()
            if u < 0.5:
                pass
            elif u < 0.8:
                new_pop = new_pop + 1
            else:
                new_pop = new_pop + 2
        pop = new_pop
        if pop == 0:
            break
    return pop
```

- (a) Compute the mean offspring $\mu = \mathbb{E}[X]$ and determine whether this branching process is subcritical, critical, or supercritical.

[2 marks]

- (b) Find the PGF $G(z)$ of the offspring distribution and compute $\text{Var}(X)$ using $G''(1) + G'(1) - [G'(1)]^2$.

[3 marks]

- (c) Write the equation $G(q) = q$ for the extinction probability q and solve it. What is the extinction probability?

[3 marks]

- (d) Compute $\mathbb{E}[Z_2]$, the expected population size at generation 2.

[2 marks]

Notes about Python:

- `random()` generates a $\mathcal{U}(0, 1)$ random variable.
- `for _ in range(n):` repeats the indented code n times (the `_` means there is no named loop variable).
- `break` exits the loop immediately.
- `pass` does nothing (the program continues to the next line).

Solution:

(a) $\mu = 0(0.5) + 1(0.3) + 2(0.2) = 0.7.$

Since $\mu = 0.7 < 1$, the process is **subcritical**.

(b) $G(z) = 0.5 + 0.3z + 0.2z^2.$

$G'(z) = 0.3 + 0.4z$, so $G'(1) = 0.7 = \mu.$ ✓

$G''(z) = 0.4$, so $G''(1) = 0.4.$

$\text{Var}(X) = G''(1) + G'(1) - [G'(1)]^2 = 0.4 + 0.7 - 0.49 = 0.61.$

(c) $G(q) = q$: $0.5 + 0.3q + 0.2q^2 = q$, i.e. $0.2q^2 - 0.7q + 0.5 = 0.$

Multiply by 5: $q^2 - 3.5q + 2.5 = 0.$

$$q = \frac{3.5 \pm \sqrt{12.25 - 10}}{2} = \frac{3.5 \pm \sqrt{2.25}}{2} = \frac{3.5 \pm 1.5}{2}.$$

So $q = 2.5$ or $q = 1.$

The extinction probability is the smallest non-negative root in $[0, 1]$, which is $q = 1.$

This confirms that subcritical processes go extinct with certainty.

(d) $\mathbb{E}[Z_n] = \mu^n.$ So $\mathbb{E}[Z_2] = (0.7)^2 = 0.49.$

Table of Distributions

Discrete Distributions $(q = 1 - p)$

Distribution	PMF $p(x)$	Support	$\mathbb{E}X$	$\mathbb{V}\text{ar}(X)$	PGF $G_X(z)$
DU($\{1, \dots, N\}$)	$\frac{1}{N}$	$\{1, \dots, N\}$	$\frac{N+1}{2}$	$\frac{N^2-1}{12}$	$\frac{z(1-z^N)}{N(1-z)}$
Ber(p)	$p^x q^{1-x}$	$\{0, 1\}$	p	pq	$q + pz$
Bin(n, p)	$\binom{n}{x} p^x q^{n-x}$	$\{0, \dots, n\}$	np	npq	$(q + pz)^n$
Geom(p)	$p q^{x-1}$	$\{1, 2, \dots\}$	$\frac{1}{p}$	$\frac{q}{p^2}$	$\frac{pz}{1-qz}$
NegBin(r, p)	$\binom{x-1}{r-1} p^r q^{x-r}$	$\{r, r+1, \dots\}$	$\frac{r}{p}$	$\frac{rq}{p^2}$	$\left(\frac{pz}{1-qz}\right)^r$
Poi(λ)	$\frac{e^{-\lambda} \lambda^x}{x!}$	$\{0, 1, 2, \dots\}$	λ	λ	$e^{\lambda(z-1)}$
Hyp(N, K, n)	$\frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}$	$\{\max(0, n+K-N), \dots, \min(n, K)\}$	$\frac{nK}{N}$	$\frac{nK(N-K)(N-n)}{N^2(N-1)}$	—

Continuous Distributions

Distribution	PDF $f(x)$	Support	$\mathbb{E}X$	$\mathbb{V}\text{ar}(X)$	MGF $M_X(t)$
$\mathcal{U}(a, b)$	$\frac{1}{b-a}$	$[a, b]$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\frac{e^{tb} - e^{ta}}{t(b-a)}$
Exp(λ)	$\lambda e^{-\lambda x}$	$[0, \infty)$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$\frac{\lambda}{\lambda - t}$
$\mathcal{N}(\mu, \sigma^2)$	$\frac{1}{\sigma \sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$	$\mathbb{R}$	μ	σ^2	$e^{\mu t + \sigma^2 t^2/2}$
Gamma(α, λ)	$\frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}$	$(0, \infty)$	$\frac{\alpha}{\lambda}$	$\frac{\alpha}{\lambda^2}$	$\left(\frac{\lambda}{\lambda - t}\right)^\alpha$
$\chi^2(n)$	$\frac{x^{n/2-1} e^{-x/2}}{2^{n/2} \Gamma(n/2)}$	$(0, \infty)$	n	$2n$	$(1-2t)^{-n/2}$
Beta(α, β)	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}$	$(0, 1)$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	—
Pareto(α)	$\alpha x^{-(\alpha+1)}$	$[1, \infty)$	$\frac{\alpha}{\alpha-1} \ (\alpha > 1)$	$\frac{\alpha}{(\alpha-1)^2(\alpha-2)} \ (\alpha > 2)$	—